

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B Tech (Electronics & Communication) Semester-III Backlog Examination May-2011

EC – 201: Digital Electronics & Logic Design

Date: 17.05.11, Tuesday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

- Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

Section I

- Q-1
- | | | |
|-----|--|---|
| (a) | Define the following terms :Power dissipation, Fan in | 2 |
| (b) | Write first 15 decimal digits in base 2 | 3 |
| (c) | Convert following number in to decimal $(1010.011)_2$ | 2 |
| (d) | Subtract the following decimal number using 9's complement. $72532-3250$ | 3 |
| (e) | Find 10's complement of $(935)_{10}$ | 1 |

- Q-2
- | | | |
|-----|--|---|
| (a) | Simplify the following Boolean function to minimum number of literals using Boolean algebra
1) $Y(WZ'+WZ)+XY$
2) $XYZ+X'Y+XY'Z$ | 4 |
| (b) | Convert the following number in other canonical form
1) $F(x,y,z)=\sum(1,3,7)$
2) $F(A,B,C,D)=\prod(0,1,2,3,4,6,12)$ | 4 |
| (c) | Design a combinational circuit to check for odd parity of three bits. A logic 1 output is required when the three bits do not constitute an odd parity | 4 |

OR

- Q-2
- | | | |
|-----|---|---|
| (a) | Simplify the following Boolean function using K map
$F(X,Y,Z)=\sum(0,2,4,5,6)$ | 3 |
| (b) | Simplify the following Boolean function using the tabulation method
$F=\sum(0,1,2,8,10,11,14,15)$ | 6 |
| (c) | Obtain the simplified expression in sum of product of following function USING K map. $X'Z'+Y'Z'+YZ'+XYZ$ | 3 |

- Q-3
- | | | |
|-----|---|---|
| (a) | Express following function in sum of min terms:
1) $F(A,B,C,D)=(A+B'+D)(A'+B+D)(C+D)(C'+D')$
2) $F(A,B,C)=(AB+C)(B+AC)$ | 4 |
| (b) | Implement the following function with a multiplexer $F(A,B,C)=\sum(1,3,5,6)$ | 4 |
| (c) | Implement a full adder circuit with decoder and two OR gates | 4 |

OR

- Q-3
- | | | |
|-----|---|---|
| (a) | Draw logical diagram of following Boolean function only with NAND gates (with two inputs) | 4 |
|-----|---|---|

- (b) $(A+D)(A'+B)(A'+C')$
Simplify the following Boolean function using don't care condition in sum of product (K map) 2
 $F=ACE+A'CD'E'+A'C'DE$
 $d=DE'+A'D'E+AD'E'$
- (c) Implement a full subtractor with two half subtractor and an or gate 4
 (d) Explain half adder and half subtractor 2

Section II

Q-4

- (a) What is the difference between sequential circuit and combinational circuit? 2
 (b) Define : binary counter, state equation 2
 (c) Explain universal shift register 3
 (d) Give the difference between synchronous and asynchronous counter 2
 (e) What is raised around condition in j-k flipflop 2

Q-5

- (a) Design a circuit that compare two 2 bit numbers A and B, to check if they are equal. The circuit has one output x, so that $x=1$ if $a=b$, $x=0$ if $a \neq b$ 4
 (b) Design a sequential circuit with four flip flops A, B, C, D. The next states of B, C and D are equal to the present state of A, B and C, respectively. The next state of A is equal to the exclusive OR of the present state of C and D. 4
 (c) Define following terms : bit time, parallel load, bidirectional shift, shift register 4

OR

Q-5

- (a) Explain 4 bit binary ripple counter with timing diagram 6
 (b) Design a counter that has a repeated sequence of six states as listed in below table 6

Excitation Table								
Count sequence			Flip flop inputs					
A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	X	0	X	1	X
0	0	1	0	X	1	X	X	1
0	1	0	1	X	X	1	0	X
1	0	0	X	0	0	X	1	X
1	0	1	X	0	1	X	X	1
1	1	0	X	1	X	1	0	X

Q-6

- Attempt any three 12
 (a) Write the difference between serial and parallel transfer?
 Explain serial transfer using example.
 (b) Show logic diagram of a clocked D flip flop with AND and NOR gates.
 (c) Write technical note on look ahead carry generator.
 (d) Draw the logic diagram (showing all gates) of a master slave D flip flop. Use NAND gates.
 (e) Write technical note on Johnson counter.